

Truffles: The Hidden Treasure Underground

An underground fungus that relies on animals and trees to survive

Phylum: Ascomycota

Tree roots

Ectomycorrhiza
(fungus + root
association)



Peridium
(outer layer)

Gleba (internal
tissue containing
asci and
ascospores)

Root → Truffle
Photosynthates (carbon)
Truffle → Root: Water +
minerals (Nitrogen/Phosphorus)

Key Concepts

- Truffles are hypogeous ascomycete fungi that develop below ground
- They form obligate ectomycorrhizal associations with host trees
- Spore dispersal is animal-mediated rather than wind-driven
- Volatile organic compounds (aroma) function to attract animals for dispersal

Animals disperse spores



ACTUAL
SIZE OF
TRUFFLE

